

PROJECT PROPOSAL

The Agrivoltaic Optimizer: Spatial Intelligence for Dual-Use Land Allocation

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1. Problem Statement

What is the "Spatial Problem"?

The global renewable energy transition faces a critical zero-sum bottleneck: large-scale solar farms threaten food security by consuming fertile arable land. The Agrivoltaic Optimizer solves this by identifying the optimal 1% of spatial zones where solar infrastructure and agriculture can mutually benefit, reducing soil evaporation while simultaneously generating power, by leveraging overlapping environmental, topographical, and infrastructure parameters with Jodhpur, Rajasthan selected as the primary study area due to its high solar irradiance potential and existing agricultural land pressure.

Target User:

Renewable energy developers, Agri-FinTech land acquisition teams, and institutional investors requiring mathematically rigorous spatial justification for ESG-compliant land leases.

2. Technical Stack & Libraries

GUI Framework:

PyQGIS - Custom interface embedded directly within QGIS to leverage native, high-performance spatial rendering.

Core Geospatial Logic:

- `geopandas` - vector data manipulation
- `psycopg2-binary` / `sqlalchemy` - PostgreSQL/PostGIS integration

The "Advanced" Component

ML & MCDA (The Core Logic):

- Multi-Criteria Decision Analysis (MCDA): Evaluates and weights overlapping criteria - solar irradiance, terrain slope, land use constraints, and proximity to electrical infrastructure - to generate a composite suitability score.
 - Machine Learning Pipeline: Primarily employs unsupervised K-Means clustering (scikit-learn) in conjunction with MCDA. Random Forest classification is architected as a future roadmap feature once labeled datasets are procured.
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Image/Raster:

- `rasterio` and `numpy` - high-speed DEM manipulation and final pixel-by-pixel suitability GeoTIFF rendering.

Database Architecture:

PostgreSQL with PostGIS (spatial vector data) and TimescaleDB (high-velocity GHI time-series logs) - a single, unified, highly optimized environment.

Data Sources:

- **Topography (DEM):** SRTM 30m tiles covering Jodhpur district
- **Infrastructure (OSM):** Grid infrastructure and road networks
- **Climate (GHI): NASA POWER** point/grid data for Jodhpur coordinates (~26.3°N, 73.0°E)
- **Land Use / Crop Cover (LULC):** Identifies agricultural zones and masks urban land, forests, and water bodies.
 - ESA WorldCover
 - Copernicus Land Cover
 - India Bhuvan LULC

3. Proposed GUI Architecture

Input Section:

A custom PyQGIS dockable panel featuring user-defined threshold sliders:

- Minimum solar irradiance limits
- Maximum distance to electrical grid
- Crop-specific shade tolerance metrics

Processing Section:

Activating the "Execute MCDA" trigger sends threshold parameters to the Python/ML backend, which queries the PostgreSQL database, applies the MCDA/K-Means algorithm, and rapidly recalculates suitability scores.

Output/Visualization:

A dynamically generated, color-coded GeoTIFF overlay rendered directly onto the QGIS map canvas, utilizing a high-contrast cartographic style to instantly highlight prime development parcels. The QGIS map canvas is pre-loaded with the Jodhpur district boundary as the base extent, with all raster outputs clipped to this AOI (Area of Interest).

4. Preliminary Task Distribution

Member Name	Primary Responsibility	Secondary Responsibility
Himanshu Bhanja	Data & Database Architecture (PostgreSQL/PostGIS/TimescaleDB)	Data Procurement (LULC, DEM, OSM, GHI) & Ingestion Pipelines
K. Yaswanthi	Lead ML & MCDA Architect	Spatial Array Normalization & K-Means Clustering Logic
R. Sathwik	PyQGIS Spatial Interface Developer	Cartographic Styling & UI/UX Event Handling

--- End of Proposal ---